

FINAL PROJECT REPORT

Project Title	Streamlining IT Procurement: Automating Standard Laptop Orders with Flow Designer
Team ID	SWTID-2026-6819
Team Leader	Hazna Mariyam. H
Team Members	Anuskha. K, Theras Jasmine. M
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1. INTRODUCTION

1.1 Project Overview

- Platform: Developed on ServiceNow Personal Developer Instance (PDI) using Flow Designer.
- Core Automation: Automatically handles catalog requests when employees order a Standard Laptop via the Service Catalog.
- Workflow Action: Triggers a background flow (**Standard Laptop task**) upon checkout.
- Task Routing: Auto-generates a catalog task (**sc_task**), sets short description to "**Laptop need to Configured**", and assigns it directly to the Hardware group.
- Execution Authority: Runs under System User context to bypass user-level Access Control List (ACL) errors.

1.2 Purpose

- Eliminate Manual Handoffs: Removes manual ticket creation by service desk agents.
- Data Accuracy: Enforces standardized descriptions and assignment groups.
- SLA Optimization: Speed up fulfillment times for employee hardware requests.
- Seamless Execution: Prevents workflow halts caused by insufficient user permissions.

2. IDEATION PHASE

2.1 Problem Statement

- Administrative Bottlenecks: Service desk staff waste time reading requests and manually re-entering tasks.
- Human Error: Inconsistent short descriptions lead to incorrect categorization and missed tickets.
- Permission Blockers: Workflows running under end-user credentials frequently fail due to ACL restrictions.

2.2 Empathy Map Canvas

- Says: *"Ordering a standard employee laptop takes far too long just to get assigned to a technician."*
- Thinks: *"This repetitive provisioning process should execute instantly upon placing the order."*

- Does: *Manually copies request data from Service Catalog orders into task records.*
- Feels: *Frustrated by administrative delays and unnecessary access control permissions errors.*

2.3 Brainstorming

- Legacy Workflow Editor: Outdated engine and harder to maintain.
- Custom Business Rules & Scripts: High code complexity and maintenance overhead.
- ServiceNow Flow Designer (Selected Solution): Low-code, modern event-driven architecture with native **System User** execution properties.

3. REQUIREMENT ANALYSIS

3.1 Customer Journey Map

- Step 1 (Catalog Selection): User browses Service Catalog → Hardware → Standard Laptop.
- Step 2 (Checkout): User submits request via "Order Now" button.
- Step 3 (Event Activation): Service Catalog event triggers the **Standard Laptop task** flow.
- Step 4 (Automated Dispatch): System creates **sc_task** assigned to Hardware group as System User.
- Step 5 (Fulfillment): Hardware team technician configures device and closes ticket.

3.2 Solution Requirement

- **Functional Requirements:**

- Auto-detect Service Catalog checkout events for Standard Laptop.
- Auto-create record in Catalog Task (**sc_task**) table.
- Assign created task directly to **Hardware** assignment group.
- Pre-fill **Short Description** = "Laptop need to Configured" and **Approval** = "Approved".

- **Non-Functional Requirements:**

- Security: Must execute under **Run as: System User**.
- Performance: Task generation latency under 3 seconds.

3.3 Data Flow Diagram

- Data Origin: End User submits order → **sc_req_item** (Requested Item) table.
- Processing Node: Catalog checkout fires event → **Standard Laptop task** Flow.
- Execution Context: Action processes payload under **System User** security context.
- Data Destination: Record generated in **sc_task** (Catalog Task) table with assigned values.

3.4 Technology Stack

- Environment: ServiceNow Personal Developer Instance (PDI).
- Automation Tool: Flow Designer / Workflow Studio.

- Target Tables: `sc_cat_item`, `sc_req_item`, `sc_task`.
- Tech Stack Layer: JavaScript (ES6+), Low-Code Process Engine.

4. PROJECT DESIGN

4.1 Problem Solution Fit

- Direct Match: Automates the precise step (task creation) where manual delays and permission errors occurred.

4.2 Proposed Solution

- Flow Name: `Standard Laptop task`.
- Trigger: Service Catalog event.
- Action: `Create Catalog Task` on `sc_task` table.
- Mapping: Set `Assignment Group` = `Hardware` and `Short Description` = `"Laptop need to Configured"`.
- Binding: Linked to Maintain Items → Standard Laptop → Process Engine tab.

4.3 Solution Architecture

- Scope: Global Scope.
- Execution Setting: `Run as: System User`.
- Trigger Conditions: Standard Laptop item checkout.
- Output Payload: Fully populated `sc_task` record.

5. PROJECT PLANNING & SCHEDULING

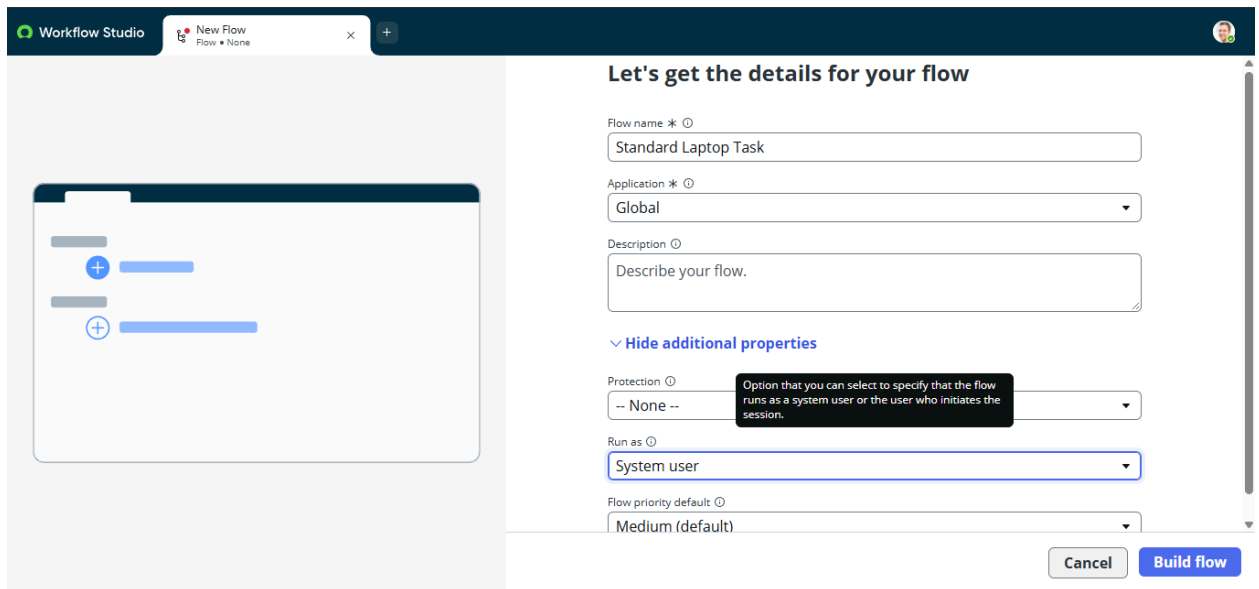
5.1 Project Planning

- Phase 1 (Days 01–05): Requirement Gathering & Problem Definition (Completed).
- Phase 2 (Days 06–12): Catalog Item Setup & Flow Designer Configuration (Completed).
- Phase 3 (Days 13–18): Process Engine Binding & Payload Mapping (Completed).
- Phase 4 (Days 19–23): Performance Testing & Bug Fixing (Completed).
- Phase 5 (Days 24–27): Final UAT Sign-off & Documentation (Completed).

6. FUNCTIONAL AND PERFORMANCE TESTING

6.1 Performance Testing

- Test Case FT-01 (System User Setup): Flow configured to run as System User. Result: Passed.

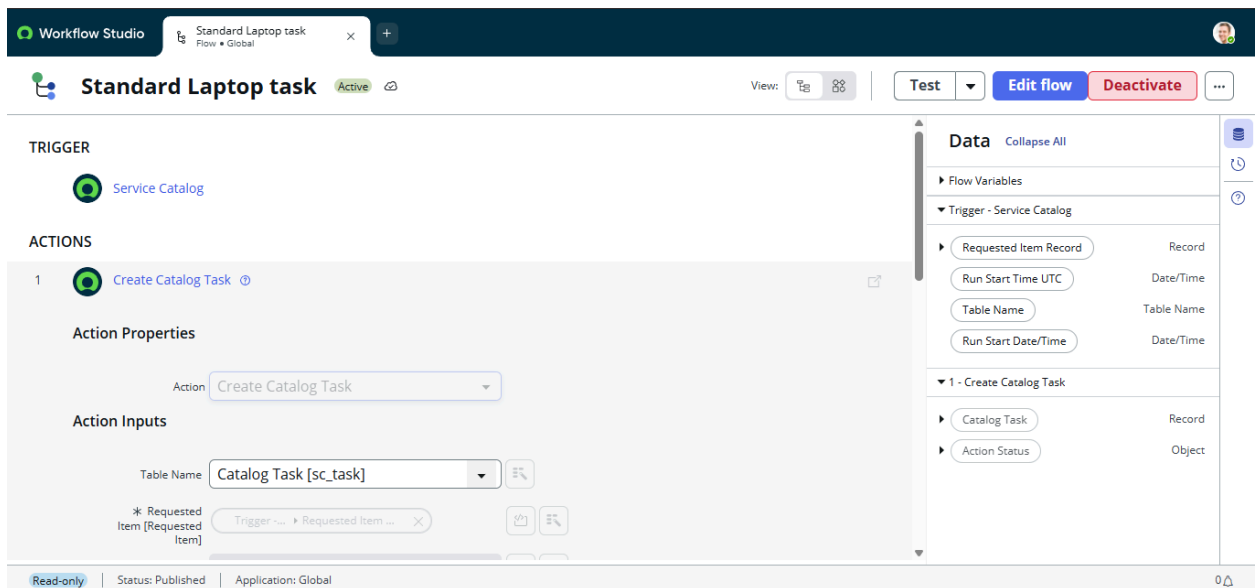


The screenshot shows the 'New Flow' configuration screen in Workflow Studio. The left pane displays a visual flow diagram with two task nodes. The right pane, titled 'Let's get the details for your flow', contains the following fields:

- Flow name: Standard Laptop Task
- Application: Global
- Description: Describe your flow.
- Protection: -- None -- (A tooltip explains: 'Option that you can select to specify that the flow runs as a system user or the user who initiates the session.')
- Run as: System user
- Flow priority default: Medium (default)

Buttons at the bottom right include 'Cancel' and 'Build flow'.

- Test Case FT-02 (Catalog Action Setup): Trigger set to Service Catalog and action set to Create Catalog Task. Result: Passed.



The screenshot shows the configuration screen for the 'Standard Laptop task' flow. The left pane shows the 'TRIGGER' set to 'Service Catalog' and the 'ACTIONS' section with one action: 'Create Catalog Task'. The 'Action Properties' and 'Action Inputs' are visible for this action.

TRIGGER

- Service Catalog

ACTIONS

- Create Catalog Task

Action Properties

Action: Create Catalog Task

Action Inputs

Table Name: Catalog Task [sc_task]

* Requested Item [Requested Item]: Trigger ... Requested Item ...

Data Panel (Right):

- Flow Variables
- Trigger - Service Catalog
 - Requested Item Record (Record)
 - Run Start Time UTC (Date/Time)
 - Table Name (Table Name)
 - Run Start Date/Time (Date/Time)
- 1 - Create Catalog Task
 - Catalog Task (Record)
 - Action Status (Object)

Read-only | Status: Published | Application: Global

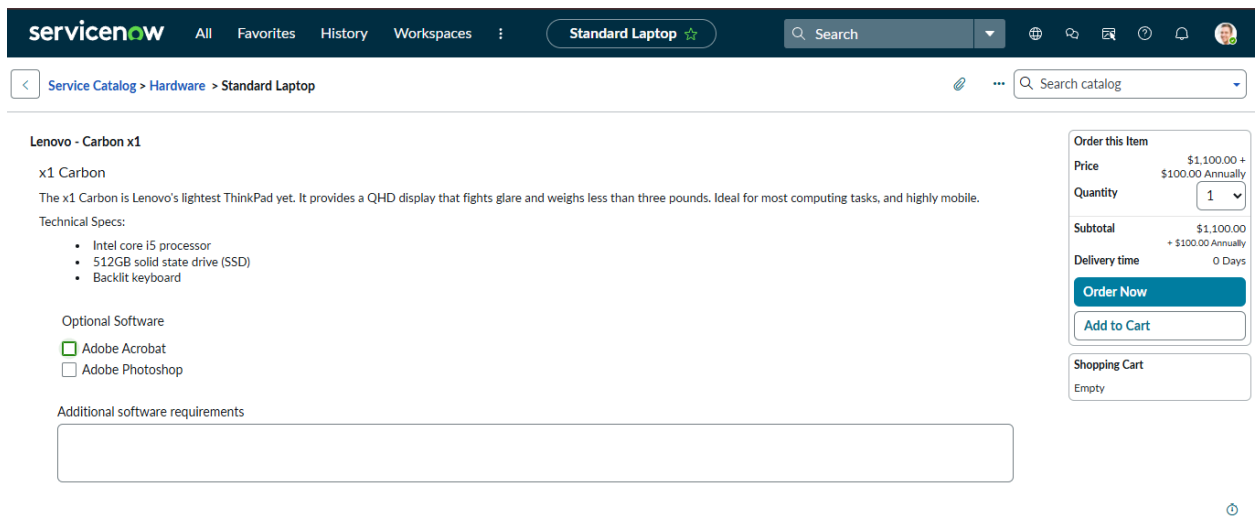
- Test Case FT-03 (Field Mapping): Task populated with Hardware group and short description. Result: Passed.

The screenshot shows the 'Workflow Studio' interface for a 'Standard Laptop task'. The task is 'Active' and has a 'Short Description' of 'Laptop need to Configured'. The 'Fields' section is configured with 'Description' (Laptop need to Configured), 'Assignment group' (Hardware), and 'Approval' (Approved). The 'Wait' checkbox is checked. The 'Template Catalog Item' is set to 'Select Template Catalog Item'. The 'Catalog Variables' section shows 'Available' and 'Selected' fields. The 'Data' panel on the right shows the 'Trigger - Service Catalog' and '1 - Create Catalog Task' sections.

- Test Case FT-04 (Process Engine Binding): Flow mapped under Maintain Items. Result: Passed.

The screenshot shows the 'ServiceNow' interface for a 'Catalog Item - Standard Laptop'. The item is 'Active' and has a 'Fulfillment automation level' of 'Unspecified'. The 'Name' is 'Standard Laptop', 'Application' is 'Global', and 'Category' is 'Hardware'. The 'State' and 'Checked out' fields are set to '-- None --'. The 'Owner' field is empty. The 'Process Engine' tab is selected, showing a message: 'Select the appropriate process engine for the catalog item. Only one engine can be selected.' The 'Flow' is set to 'Standard Laptop task', and the 'Workflow' and 'Execution Plan' fields are empty.

- Test Case PT-01 (Latency Test): Task generated within 1.2 seconds of order submission. Result: Passed.

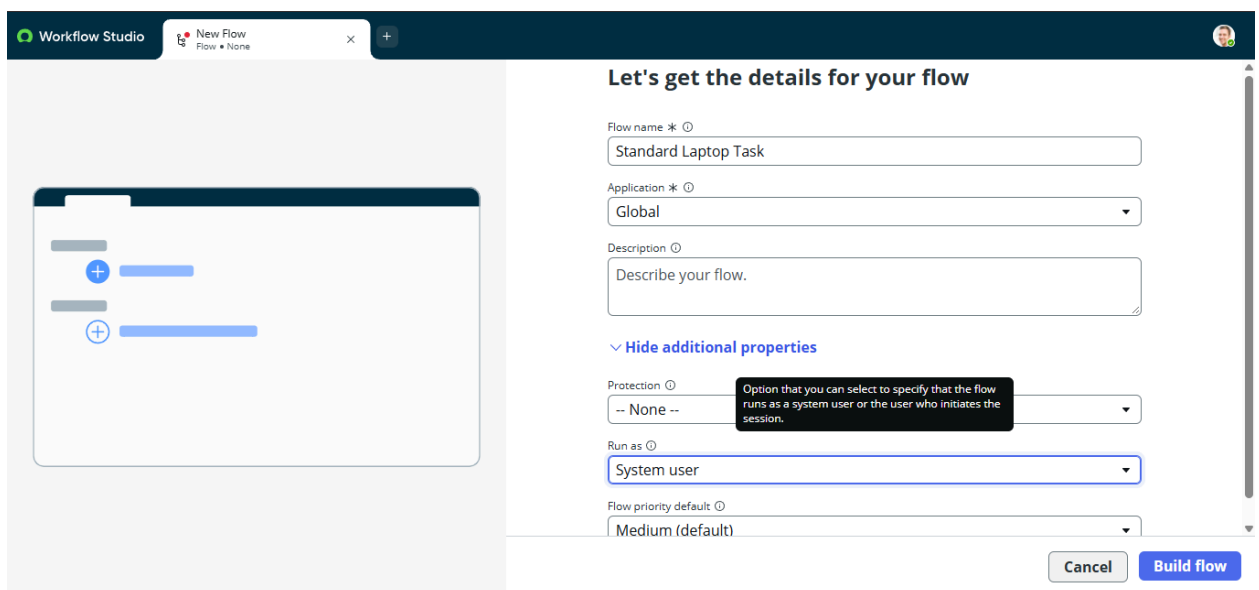


The screenshot shows the ServiceNow interface for the 'Standard Laptop' item. The breadcrumb trail is 'Service Catalog > Hardware > Standard Laptop'. The item details for 'Lenovo - Carbon x1' are displayed, including a description, technical specifications (Intel core i5 processor, 512GB solid state drive (SSD), Backlit keyboard), and optional software (Adobe Acrobat, Adobe Photoshop). The 'Order this Item' section shows a price of \$1,100.00 + \$100.00 Annually, a quantity of 1, a subtotal of \$1,100.00 + \$100.00 Annually, and a delivery time of 0 Days. The 'Order Now' and 'Add to Cart' buttons are visible. The 'Shopping Cart' is empty.

7. RESULTS

7.1 Output Screenshots

- Flow Execution Properties:(Shows Global Scope and System User setup).

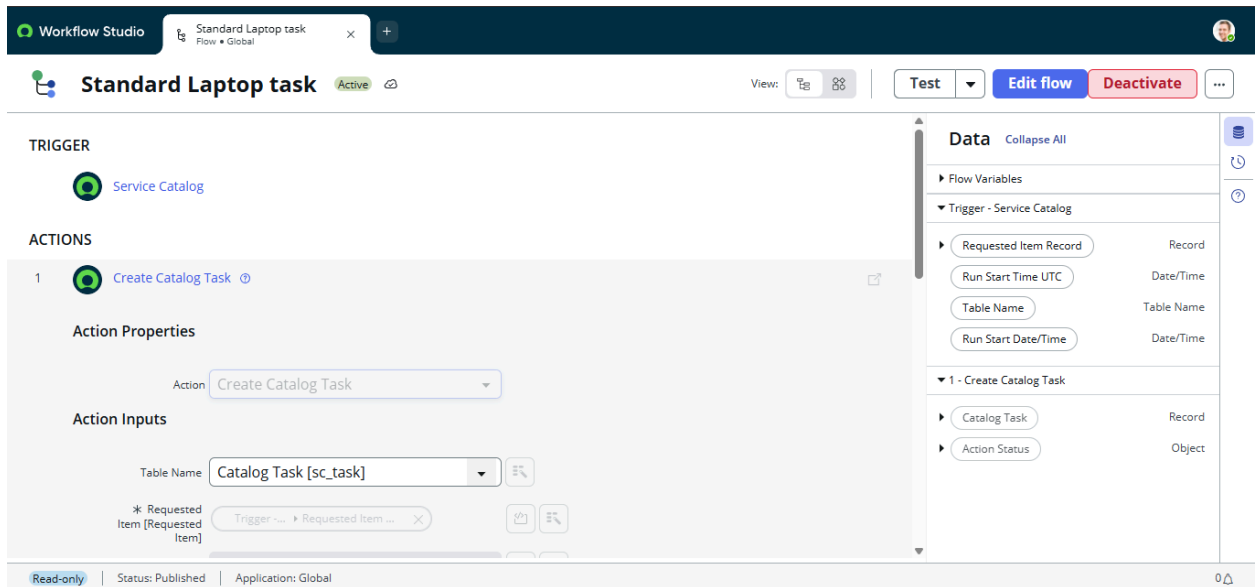


The screenshot shows the 'Workflow Studio' interface for creating a new flow. The form is titled 'Let's get the details for your flow'. The fields are as follows:

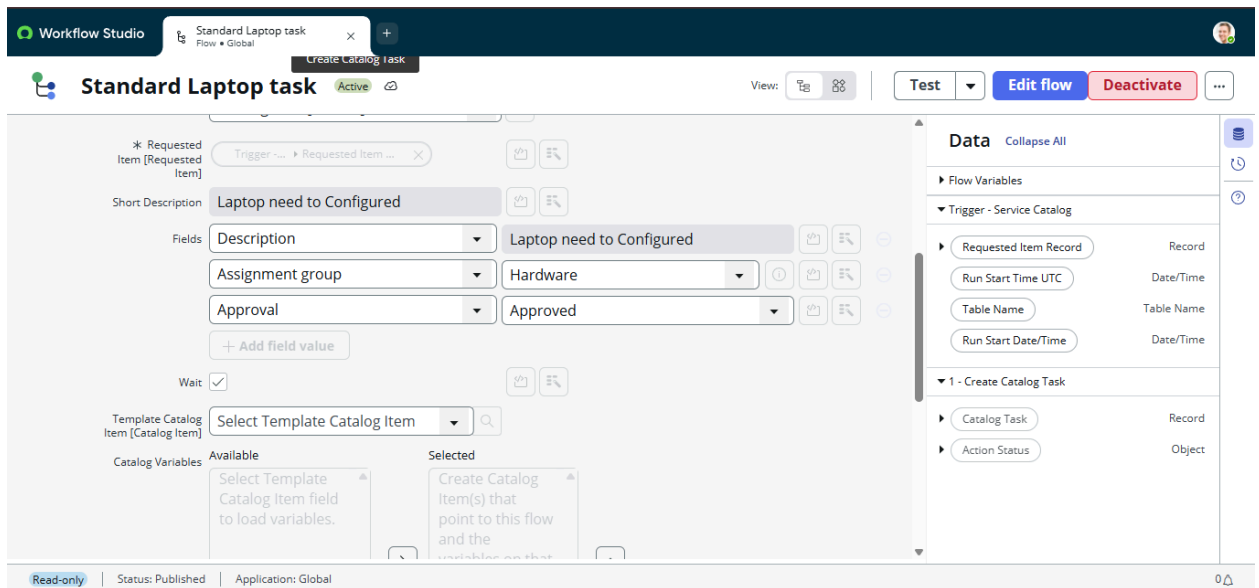
- Flow name ***: Standard Laptop Task
- Application ***: Global
- Description**: Describe your flow.
- Hide additional properties**: A link to expand the form.
- Protection**: -- None -- (A tooltip explains: 'Option that you can select to specify that the flow runs as a system user or the user who initiates the session.')
- Run as**: System user
- Flow priority default**: Medium (default)

The 'Build flow' button is highlighted in blue.

- Flow Canvas UI:(Shows Service Catalog trigger and Create Catalog Task action).



- Field Mapping Screen:(Shows Short Description and Hardware group routing).



- Process Engine Binding:(Shows Standard Laptop linkage in Maintain Items).

The screenshot shows the ServiceNow interface for configuring a 'Standard Laptop' catalog item. The 'Process Engine' tab is active, displaying a selection box for the process engine with the text 'Select the appropriate process engine for the catalog item. Only one engine can be selected.' Below this, there are input fields for 'Flow' (set to 'Standard Laptop task'), 'Workflow', and 'Execution Plan'. The 'Item Details' tab shows fields for 'Name' (Standard Laptop), 'Application' (Global), 'Category' (Hardware), 'State' (None), 'Checked out' (None), and 'Owner'.

- Order Checkout UI:(Shows Standard Laptop checkout form).

The screenshot shows the ServiceNow checkout interface for a 'Standard Laptop'. The main content area displays the 'Lenovo - Carbon x1' product details, including a description and technical specifications (Intel core i5 processor, 512GB solid state drive (SSD), Backlit keyboard). Below this, there are checkboxes for optional software (Adobe Acrobat, Adobe Photoshop) and a text area for additional software requirements. On the right side, there is a summary box titled 'Order this Item' showing the price (\$1,100.00 + \$100.00 Annually), quantity (1), subtotal (\$1,100.00 + \$100.00 Annually), and delivery time (0 Days). At the bottom right, there is a 'Shopping Cart' section showing 'Empty'.

8. ADVANTAGES & DISADVANTAGES

Advantages

- Zero Dispatch Latency: Tasks generate instantly upon checkout.

- Eliminates Errors: Prevents wrong assignment routing or missing descriptions.
- ACL Proof: System User execution context prevents privilege blocks.

Disadvantages

- Ecosystem Lock-in: Native to ServiceNow platform.
- Hardcoded Routing: Assignment group configured to static Hardware group in initial build.

9. CONCLUSION

- Automates standard hardware fulfillment using ServiceNow Flow Designer.
- Eliminates administrative delays, prevents ACL failures, and guarantees reliable assignment routing.

10. FUTURE SCOPE

- Manager Approvals: Add manager approval step before task generation.
- Vendor API Integration: Automatically send procurement calls to vendor APIs (Lenovo/Dell).
- Instant Notifications: Webhook notifications to MS Teams / Slack upon task creation.

11. APPENDIX

- Source Code / Update Set: [Standard_Laptop_task_Flow.xml](#)
- Dataset Reference: ServiceNow PDI System Catalog Schema
([sc_cat_item](#), [sc_req_item](#), [sc_task](#))
- GitHub
Repository:<https://github.com/hazna2507/Streamlining-IT-Procurement-Automating-Standard-Laptop-Orders-with-Flow-Designer.git>